DPP	DEPARTMENTAL POLICY AND PROCEDURE		
Title	**CARE OF ENDOTRACHEAL TUBE**		
Department	**Neonatology Department**		
Applies to	**All NICU Medical and Nursing Staff, Respiratory Therapist**		
Policy Code and Version Number	**DPP-NICU-007 (v1)**		
Last Version Approval Date	**N/A**	Number of Pages	**5**

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APPROVAL DATE	EFFECTIVE DATE	TYPE OF REVISION	REVIEW DATE
19/3/2024	7/4/2024	TRIENNIAL	18/3/2027

1.0 PURPOSE

1.1. The procedure will facilitate removal of secretions;

1.2. To relieve CO2 retention in clients with chronic pulmonary disease and treats acute respiratory failure.

1.3. To maintain a patent airway.

2.0 DEFINITION/ABBREVIATION

2.1. **Endotracheal tube** is a flexible tube inserted through the mouth or nose and into the trachea beyond the vocal cords that acts as an artificial airway. It will maintain a patent airway, allows for deep tracheal suction and removal of secretions, and permits mechanical ventilation.

2.2. **NICU** – Neonatal Intensive Care unit

2.3. **Sa02** – Arterial Oxygen saturation

2.4. **ETT** – Endotracheal tube

2.5. **RT** – Respiratory Therapist

2.6. **NGT** – Nasogastric tube

3.0 RESPONSIBILITY

3.1. **Head nurse** has to make sure that all needed equipments are available and staffs are competent to perform the procedure.

3.2. **Charge nurses** must supervise the bedside nurses and oversee the procedure. Everyone has to comply with the policies.

3.3. **Staff nurses** will be responsible for continuous monitoring of ETT patency, inform RT for any changes with the settings and to troubleshoot in equipment malfunction and direct care of patient like suctioning with the help of RT.

3.4. **Respiratory therapist** has to make a frequent rounds and assessment to patient and double check ventilatory settings.

4.0 POLICY

4.1. Nurses must be knowledgeable in caring patient with ETT and connected to mechanical ventilator.

4.2. Indications.

4.2.1. Provide airway for mechanical ventilatory support.

4.2.2. Relieve critical upper airway obstruction.

4.2.3. Provide route for selective bronchial ventilation.

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4.2.4. Assist in pulmonary hygiene when secretions cannot be otherwise cleared.

4.2.5. Obtain direct tracheal cultures.

4.3. The correct endotracheal tube (ETT) size and length of insertion (tip to lip distance) can be estimated from the infant's weight.

Weight	ETT	Depth of Insertion (cm)
1 kg	2.5	7
2 kg	3.0	8
3 kg	3.5	9
4 kg	4.0	Add 1 cm for each additional kg of body weight
		Insertion depth (cm) = cm + wt (kg)

The tube should not fit tightly between the vocal cords in order to minimize upper airway trauma

4.4. Nurses have to make sure that ventilator settings are correlating with the prescribed physicians order.

4.5. Keep the head of the bed elevated 30 – 40 degrees at all times if patient conditions allows.

4.6. High levels of inspiratory pressure with PEEP increase the risk of barotraumas and pneumotorax, regular assessment of breath sounds and oxygenation status is important.

4.7. Ventilated patient must be well nourished; an NGT must be intact as per physician's order.

4.8. Caring for a patient on mechanical ventilator requires teamwork, knowledge of care goals and interventions based on best practices, patient needs and response to therapy.

4.9. Daily assessment of patient using the infection control bundle checklist.

5.0 PROCEDURE (S)

5.1. Gather supplies for care of endotracheal tube.

5.2. Securing of ETT is across the maxilla (upper jaw) of child to decrease chance of displacement, unless patient condition contraindicates this.

5.3. Securing of Endotracheal Tubes requires RT and nurse. A qualified physician in intubation must be readily available when securing ETT. When ETT unsecured, one personnel must always hold the ETT until securing is completed.

5.4. Note the proper ETT level of insertion (depth) in cm.

5.5. While changing plaster, hold tube securely. If using adhesive tape method remove any old adhesive with adhesive tape remover and wipe off with wet face cloth

5.6. ETT should be repositioned on a PRN basis

5.7. Patient monitoring is required during ETT care and securing such as carbon dioxide monitoring and SPO2.

5.8. Prepare for analgesic and sedation according to the physician's order before attempting to re-change plasters.

5.9. Cleanse cheeks. Secure endotracheal tube and airway or block. Firmly secure in place. Use the upper lip to anchor the tape.

5.10. Apply mustache tape to upper lip. - Cut strip width from tape according to the size of the upper lip of the child

5.11. Cut two strips of Elastoplast tape (width and length will depend on the size of the baby) and cut down the middle of half of the tape. - Tape left or right side of the tube by putting upper strip of tape from cheek to cheek under nose. Use lower tape to secure tube, wrapping around 2 – 3 times and securing on the same side.

5.12. **Routine care of endotracheal tubes**

5.12.1. Ensure continuous monitoring of patient. ETCO2 monitoring should be used where feasible.

5.12.2. Ensure appropriate sized facemask and BVM connected to oxygen flow meter is readily available at all times.

5.12.3. Auscultate lung fields for adequate air entry for children every two hourly

5.12.4. Continuously assess patient for deterioration and complications of displacement, obstruction, pneumothorax, oxygenation and ventilation equipment failure.

5.12.5. Use oral airway or bite block if necessary.

5.12.6. Place ventilator / humidity tubing in dependent position in order to drain condensation away from patient.

5.12.7. The cuff pressure should not exceed 25 mmHg or there may be increased risk of pressure necrosis at the site where the cuff contacts the tracheal mucosa.

5.13. **Protection of the patient from self-extubation**

5.13.1. Ensure the ETT is properly secured at all times

5.13.2. Prevent pulling and jarring of the ventilator / humidity tubing and ETT.

5.13.3. Patient may need to be restrained if the nurse feels it is necessary to prevent patient from pulling at the tube.

5.13.4. Sedate patient as required per physician's orders

5.13.5. Continuously monitor for air leak: Gurgling sounds, Vocalization, Loss of tidal volume (if ventilated)

5.13.6. Patients should be suctioned using sterile technique. Suctioning the ETT must not be deep to avoid trauma, the following formula can be used to maintain the length of suctioning. Maximum Suction Catheter Depth (MSCD).

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5.13.6.1. ETT: Length from the junction of the tube + 3.5 cm

5.13.6.2. Inline suction: total length of tube + 5 cm

5.13.7. The optimal frequency of suctioning depends on the quantity of secretions or as per physician's order

5.13.8. Document. Size and length of the ETT, ventilator setting and patient response to procedure.

6.0 EQUIPMENTS

6.1. Appropriate size of bag-valve-mask
6.2. Adhesive tape
6.3. Wet and dry face cloths
6.4. Bite block with oro pharyngeal airway
6.5. Personal protective equipment

7.0 ATTACHMENT/FORMS

7.1. Nurses Progress Notes (GDOH-NUR-NPN-203)

8. REFERENCES

8.1. Ministry of Health, Policy and Procedure Manual, KSA

8.2. Lippincott Manual of Nursing Practice, 9th Edition

8.0 REVISION CLAUSE

9.1. Not applicable

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Region: _____ المنطقة/المحافظة:

Dept./Unit: _____ القسم/الوحدة:

MRN: | | | | | | | | | | رقم الملف الطبي:

Name: _____ الاسم:

Nationality: _____ الجنسية:

Age: _____ ☐ سنة Years ☐ شهر Months ☐ يوم Days العمر:

Date of Birth: ____ / ____ / 14____ H ____ / ____ / 20____ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

NURSES PROGRESS NOTES FORM

DATE &TIME	NURSES NOTES (D-data A-action R-response)	REGISTERED NURSE INITIAL WITH ID NUMBER

Note: Write the time in each entry & affix your initial at the end of each paragraph. Document your complete Name, Initial, Job number, Date & Time at the closure of your documentation. Draw a line across empty spaces.

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Name: _____ الاسم:	MRN: رقم الملف الطبي:

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